

Table 1. Overall MGSM multilingual-transfer results for Qwen3-4B, macro-averaged equally across five languages with 250 problems per language. Higher is better for every column.

	Method	Overall steering (0–2) $\uparrow$	Exact answer (%) $\uparrow$	Spanish rule (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Original	0.000	41.6	0.00	1.76	1.47
	S-PID	0.000	0.0	1.98	0.00	0.00
	A-LQR	1.270	22.4	1.81	1.51	1.08
	H_∞ (ours)	1.141	32.2	1.44	1.66	1.25